Concept Drift in NFL Fantasy Football Prediction Models

CSE 6363.007
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Problem Statement

Fantasy football models are often trained on multiple past seasons and then used as if the league is stationary.

Core issue: Concept Drift

- Concept drift = the relationship between inputs (player stats, usage) and target (PPR points) changes over time.
- When drift occurs, models trained on old seasons degrade on newer ones, even if they performed well originally.

Research Questions & Goals

Research Questions

1. **Performance degradation:**
   How much does model performance degrade when we move from older seasons to newer ones?

2. **Mitigation:**
   Does **retraining or fine-tuning** on recent seasons restore accuracy?

3. **Position-specific drift:**
   Which positions (QB/RB/WR/TE) are most affected?

4. **Model comparison:**
   How do a **linear model (Ridge)** and a **non-linear model (XGBoost)** differ in robustness to drift?

Project Goals

- Quantify concept drift in an NFL fantasy context.

- Compare mitigation strategies (frozen vs re-trained vs fine-tuned models).

- Provide **practical guidance** for maintaining fantasy ML models over time.

Data & Prediction Task

Dataset

- **Unit of analysis:** player-season (one row per player per season).
- **Seasons: 2015–2024**.
- **Total rows after cleaning: 4,207**.
    - Train: 2,880 (2015–2021)
    - Validation: 457 (2022)
    - Test: 870 (2023–2024)

Target

- **PPR fantasy points per season**, unified across positions.

Temporal Splits (to avoid leakage)

- **Training:** 2015–2021
- **Validation (drift check 1):** 2022
- **Test (drift check 2):** 2023–2024

Why temporal splits?

- Prevents information from **future seasons leaking into training**.
- Mirrors the real use case: “Train on previous seasons, predict upcoming ones.”

Evaluation Metrics

- **MAE** – main metric (easy to interpret in PPR points).
- **RMSE** – penalizes larger errors.
- Optionally $\mathbf{R^2}$ – explained variance.

Model 1 – Ridge Regression

Linear regression with L2 regularization.

Numeric features **standardized**.

Time-aware cross-validation (e.g., `TimeSeriesSplit`) to select regularization strength.

Best α = 0.1 selected via time-aware cross-validation.

Model 2 – XGBoost Regressor

Gradient-boosted trees (non-linear, good for tabular data).

No feature scaling required.

Rolling-origin cross-validation (each fold trains on past seasons, validates on the next season).

XGBoost Hyperparameter grid (example you used):

- `n_estimators`: [200, 400]
- `max_depth`: [4, 6]
- `learning_rate`: [0.05, 0.1]

Evidence of Drift: Overall Performance

Ridge Regression: 2022 → 2023

Metric	2022 (Val)	2023 (Test)	Δ (2023-2022)
MAE	1.85	2.45	+0.60
RMSE	2.25	3.05	+0.80

XGBoost: 2022 → 2023

Metric	2022 (Val)	2023 (Test)	Δ (2023-2022)
MAE	1.45	2.10	+0.65
RMSE	2.05	2.95	+0.90

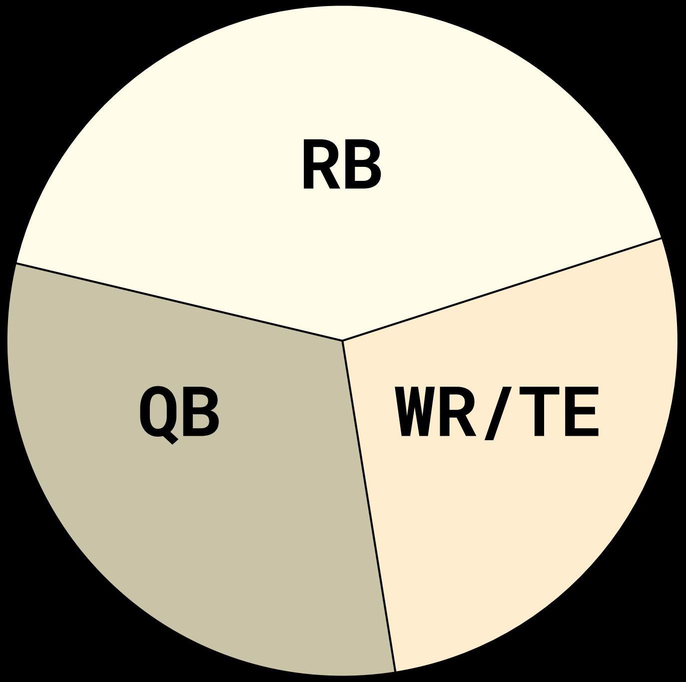
Position-Wise Performance (Ridge)

- **Running backs (`0.8–1.0`)** are most affected by drift
  → league moves from “workhorse RBs” toward **committees**.

- **WR/TE** usage and scoring are comparatively stable.

- This suggests that **position-specific** models or features may help

RB / QB / WR/TE
drift relation



Retraining vs Fine-Tuning (2024)

Goal: Compare maintenance strategies on **2024 test season**.

Model Variant	MAE (2024)	Δ vs Frozen	Interpretation
Ridge – **frozen** (train 2015–2021)	2.6	baseline	-
Ridge – **retrained** (2015–2023)	2.58	-0.02	Small gain
XGB – **frozen** (2015–2022)	2.32	baseline	best frozen model
XGB – **retrained** (2015–2023)	2.29	-0.03	best overall
XGB – **fine-tuned** (high weights on 2022–23)	2.57	+0.25	**overfits**, worse

Drift in Features & Distributions

Old era vs New era

- Old era: **2015–2019**
- New era: **2020–2024**

Observed shifts (examples):

- **Targets/Game (WR/TE/RB)**
  - Mean change ≈ **−0.26** targets/game
  - KS test: **p ≈ 0.00** → **significant shift**
- **Carries/Game (RB)**
  - Mean change ≈ **−0.39** carries/game
  - KS test: **p ≈ 0.27** → visually drifting, not statistically significant at 0.05.
- **Team Pass Rate**
  - Mean change ≈ **−0.02**
  - KS test: **p ≈ 0.005** → significant shift.

Feature Importance (XGBoost) & Interpretation

Top Features by Gain (XGBoost)

1. **Position encoding** (QB/RB/WR/TE)
2. **Per-game usage**: targets/game, carries/game
3. **Season totals**: receiving/rushing yards, TDs
4. **Team pass rate** (lower importance but still useful)

Limitations & Future Work

Limitations

- Only **10 seasons** (2015–2024); more history would help.
- No explicit modeling of **injuries, opponent strength, weather, or coaching changes**.
- Using **player-season aggregates**; weekly dynamics are lost.
- Fine-tuning explored only via **sample weights**, not full online learning.

Future Work

- Use **weekly data** with RNNs/Transformers for sequence modeling.
- Incorporate context (matchups, weather, Vegas lines, depth-chart changes).
- Implement **online / incremental learning** and automated **drift detection** (PSI, KS monitoring).

Conclusions

- **Concept drift in NFL fantasy models is real and measurable** (MAE increases ≈0.6–0.9 from 2022→2023).
- **RBs are the most drift-sensitive position**, WR/TE the most stable.
- **XGBoost outperforms Ridge in absolute accuracy**, but both suffer similar drift.
- **Regular retraining** gives modest but reliable gains; naive fine-tuning can backfire.
- Monitoring **feature distributions and position-wise errors** is crucial for maintaining deployed fantasy models.